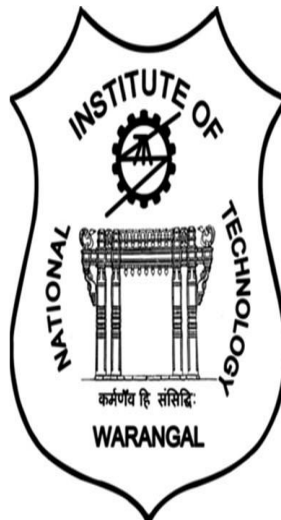


A
Minor Project report
on
SPATIAL INTERPOLATION OF TEMPERATURE AND
HUMIDITY USING DHT22 SENSOR DATA: A WEB GIS-
BASED VISUALIZATION APPROACH

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ABSTRACT

Environmental monitoring of temperature and relative humidity plays a critical role in agricultural planning, urban heat island analysis, hydrological modeling, and climate research. Traditional monitoring systems based on fixed meteorological stations are spatially limited and often fail to capture fine-scale microclimatic variations, especially within campus or urban environments. This project addresses these limitations by deploying a network of DHT22 low-cost sensors at multiple spatial locations within the NIT Warangal campus and applying spatial interpolation techniques to generate continuous environmental surface maps.

The DHT22 sensors, mounted on STM32 Nucleo-L476RG microcontrollers, record temperature (with $\pm 0.5^{\circ}\text{C}$ accuracy) and relative humidity (with $\pm 2\text{--}5\%$ accuracy) at one-minute intervals. Each data point is geo-tagged with sensor ID, geographic coordinates in WGS 84 (EPSG:4326), and a timestamp. The collected data is stored in CSV format, preprocessed for quality control, and synchronized across sensors to form a standardized geospatial dataset.

Inverse Distance Weighting (IDW) spatial interpolation is applied to convert discrete point observations into continuous raster surface maps. IDW operates on the principle that nearby observations exert greater influence than distant ones, with the power parameter controlling the rate of influence decay.

The interpolated raster outputs are integrated into a WebGIS platform built on Leaflet.js and GeoServer, visualization through zooming, layer toggling, and point-level data retrieval.

Results demonstrate significant microclimatic variability across the NIT Warangal campus. Temperature values ranged from 24°C to 33°C and humidity from 45% to 75%, with differences of $6\text{--}8^{\circ}\text{C}$ and 20–30% observed within distances of 500–800 meters. Built-up and exposed areas recorded higher temperatures above 30°C corresponding to lower humidity below 55%, while vegetated regions showed the inverse pattern — lower temperatures below 27°C and higher humidity above 65%. This negative correlation between temperature and humidity was effectively captured owing to the high spatial resolution of the sensor network.

Compared to conventional weather station networks, the DHT22-based dense sensor array provides superior spatial coverage, micro-scale variation detection, and the ability to identify localized hotspots and humidity zones. The integration of IoT sensor data with WebGIS visualization provides a scalable, low-cost, and accessible approach to environmental monitoring at local scales.

Keywords: DHT22, Spatial Interpolation, IDW, Temperature, Humidity, WebGIS, Microclimate, NIT Warangal, Leaflet.js, GeoServer, IoT, Environmental Monitoring

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
DHT22	Digital Humidity and Temperature Sensor (Model 22)
GIS	Geographic Information System
WebGIS	Web-based Geographic Information System
IDW	Inverse Distance Weighting
OK	Ordinary Kriging
IoT	Internet of Things
RMSE	Root Mean Square Error
WGS	World Geodetic System
EPSG	European Petroleum Survey Group (coordinate reference systems)
CSV	Comma-Separated Values
STM32	STMicroelectronics 32-bit ARM Microcontroller
LCD	Liquid Crystal Display
API	Application Programming Interface
GeoJSON	Geographic JavaScript Object Notation
CRS	Coordinate Reference System
DEM	Digital Elevation Model
LST	Land Surface Temperature

NDVI	Normalized Difference Vegetation Index
NIT	National Institute of Technology
GPI	Global Polynomial Interpolation
LPI	Local Polynomial Interpolation
M-IDW	Modified Inverse Distance Weighting
URL	Uniform Resource Locator
HTTP	Hypertext Transfer Protocol
OGC	Open Geospatial Consortium
WMS	Web Map Service
WFS	Web Feature Service

CHAPTER 1: INTRODUCTION

1.1 Background and Context

Temperature and relative humidity are two of the most fundamental meteorological parameters that govern a wide range of physical, biological, and environmental processes. Their spatial and temporal variation directly influences agricultural productivity, water resource management, urban climate dynamics, public health outcomes, and infrastructure planning. Accurate, high-resolution mapping of these parameters at local scales is therefore essential for informed decision-making in both engineering and environmental management contexts.

Conventional approaches to monitoring temperature and humidity rely on fixed meteorological stations that record atmospheric conditions at specific geographic points. While these stations provide high temporal resolution and generally acceptable accuracy, their spatial density is inherently limited by the costs associated with installation, maintenance, and data communication infrastructure. In a typical monitoring network, stations may be spaced several kilometers apart, making it practically impossible to capture fine-scale spatial heterogeneity within a local environment such as a university campus, an agricultural field, or an urban neighbourhood.

The emergence of low-cost, commercially available solid-state sensors has opened new possibilities for dense environmental monitoring. The DHT22 sensor, in particular, combines temperature and humidity sensing capabilities in a compact, inexpensive package that is easily interfaced with microcontroller platforms such as the STM32 Nucleo series. By deploying arrays of such sensors at multiple spatial locations, it becomes possible to collect geo-referenced environmental data at resolutions that were previously unachievable with conventional station-based networks.

However, even with dense sensor deployments, observations remain point-based by nature. The transition from discrete point measurements to continuous spatial representations requires mathematical spatial interpolation techniques. These methods use the values recorded at known locations to estimate environmental conditions at unmeasured locations, effectively generating spatially continuous surface maps from scattered point data.

Spatial interpolation has been a cornerstone of Geographic Information Science (GIScience) for several decades. Deterministic methods such as Inverse Distance Weighting (IDW) and Spline interpolation, as well as geostatistical methods such as Kriging, are widely used in environmental mapping applications. The choice of interpolation method significantly affects

the quality and accuracy of the resulting surface, and extensive research has been conducted to understand the comparative performance of these methods under different data conditions.

The visualization and dissemination of interpolated environmental maps has traditionally been confined to desktop GIS software platforms such as ESRI ArcGIS and QGIS. While these tools offer comprehensive analytical capabilities, they require technical expertise and are not accessible to general users or non-specialist stakeholders. WebGIS platforms, built on open-source technologies such as Leaflet.js and GeoServer, address this limitation by enabling interactive map visualization through standard web browsers without requiring any specialized software installation.

1.2 Problem Statement

Despite significant advances in both sensor technology and geospatial computing, a critical gap exists in the application of low-cost sensor networks for high-resolution microclimate mapping at local scales in Indian cities and institutional campuses. Existing environmental monitoring infrastructure within academic institutions such as NIT Warangal typically consists of one or at most a few meteorological instruments, which are insufficient for characterizing the spatial heterogeneity of temperature and humidity across the campus landscape.

The NIT Warangal campus encompasses a variety of land cover types including built-up areas with concrete buildings, roads, open grounds, and vegetated zones with trees and lawns. These diverse surface characteristics generate distinct microclimatic conditions that cannot be captured by a single monitoring point. Consequently, decisions related to campus greening, heat mitigation, and outdoor thermal comfort are made in the absence of spatially detailed environmental data.

Furthermore, even when sensor data is collected, the absence of accessible visualization tools means that the insights derived from spatial interpolation are typically accessible only to trained GIS professionals and not to the broader community of campus administrators, students, and planners who could benefit from such information. A web-accessible, browser-based visualization platform would democratize access to this data and support evidence-based environmental management.

1.3 Importance of Spatial Interpolation in Environmental Monitoring

Spatial interpolation is a process by which values at unmeasured locations are estimated based on values observed at a finite set of known points. This technique is fundamental to environmental science, hydrology, meteorology, and geography because it allows researchers

and practitioners to construct continuous spatial representations of phenomena that are inherently measured at discrete locations.

In the context of temperature and humidity monitoring, spatial interpolation serves several critical functions. First, it enables the creation of continuous spatial surfaces from sensor point data, allowing the full spatial extent of a study area to be characterized rather than only the specific sensor locations. Second, it facilitates the identification of spatial gradients and transitions between zones of different environmental conditions, which is essential for understanding the physical processes driving those conditions. Third, interpolated surfaces can be used as inputs to downstream models, such as urban heat island assessments, hydrological simulations, and crop stress analyses.

Among the available interpolation methods, Inverse Distance Weighting (IDW) is one of the most widely applied in practice due to its conceptual simplicity, computational efficiency, and ease of implementation. IDW operates on the assumption that the value at any unmeasured location is a weighted average of the values at surrounding measured points, with weights inversely proportional to distance raised to a specified power. The power parameter controls the degree to which nearby points dominate the estimate relative to more distant ones.

Kriging, a geostatistical interpolation method, provides an alternative that accounts for spatial autocorrelation in the dataset by modeling the variance structure of the data through a semivariogram. While Kriging generally provides more accurate estimates when the data exhibit clear spatial autocorrelation, it requires more computational effort and a sufficient number of sample points to reliably estimate the semivariogram parameters. IDW is therefore often preferred for smaller datasets or as a first-pass analysis tool.

Spline interpolation is a third category of deterministic interpolation that fits a smooth mathematical surface through the data points, similar to bending a flexible sheet of material. Spline methods are well-suited for generating smooth surfaces but may introduce artificial oscillations (the Runge phenomenon) when applied to datasets with abrupt spatial transitions.

The choice between IDW, Kriging, Spline, and other interpolation methods depends on the nature of the data, the spatial distribution of sample points, the required accuracy, and the intended application. No single method universally outperforms others, and validation through cross-validation or hold-out testing is advisable when high accuracy is critical.

1.4 Role of DHT22 Sensors in Microclimate Monitoring

The DHT22 sensor (also known as the AM2302) is a digital relative humidity and temperature sensor that has gained widespread adoption in IoT (Internet of Things) and embedded systems

applications due to its combination of affordability, accuracy, and ease of use. Key technical specifications include a temperature measurement range of -40°C to $+80^{\circ}\text{C}$ with an accuracy of $\pm 0.5^{\circ}\text{C}$, and a humidity measurement range of 0% to 100% relative humidity with an accuracy of $\pm 2\%$ to $\pm 5\%$ RH.

The sensor communicates via a single-wire digital protocol, simplifying interface requirements and reducing the number of components needed. The DHT22 provides readings at intervals of approximately 2 seconds, making it suitable for continuous monitoring applications. Its compact form factor allows for flexible deployment in indoor and semi-outdoor environments.

In this project, the DHT22 sensors are paired with the STM32 Nucleo-L476RG microcontroller platform, which provides the processing capability to read sensor data, apply timestamp tagging, store readings in memory, and transmit data via serial communication. Each sensor-microcontroller unit is further equipped with a JHDN162A 16x2 LCD display for local visual feedback, allowing on-site verification of readings without the need for a computer interface.

The use of multiple DHT22 sensors at spatially distributed locations transforms what would otherwise be a single-point measurement device into a spatial sensor network capable of capturing the environmental heterogeneity of the study area. This transformation is the technical foundation upon which the spatial interpolation and WebGIS visualization components of the project are built.

1.5 WebGIS for Environmental Data Visualization

WebGIS refers to the delivery of geographic information system functionality through web-based interfaces, enabling users to interact with maps, spatial data, and geographic analysis tools through a standard web browser. The emergence of open-source WebGIS frameworks has dramatically lowered the technical and financial barriers to deploying interactive mapping applications.

Leaflet.js is a widely used open-source JavaScript library for creating interactive web maps. It provides a lightweight, efficient, and highly customizable framework for displaying tile layers, vector data, raster overlays, and custom markers. GeoServer is a complementary open-source server platform that implements OGC (Open Geospatial Consortium) standards including Web Map Service (WMS) and Web Feature Service (WFS), enabling standardized delivery of geospatial data to web clients.

The combination of Leaflet.js on the client side and GeoServer on the server side creates a powerful architecture for deploying browser-accessible environmental monitoring applications. Users can zoom into specific areas of interest, toggle between temperature and humidity layers,

query individual sensor points for their recorded values, and visualize temporal variation of environmental parameters through time-controlled playback features.

For the purposes of this project, the WebGIS platform serves both as a data delivery mechanism and as a decision support tool. Campus administrators, environmental engineers, and students can use the platform to understand the spatial distribution of temperature and humidity across the NIT Warangal campus, identify areas of thermal stress, and monitor seasonal or diurnal variations in microclimate conditions without requiring any GIS expertise or software installation.

CHAPTER 2: LITERATURE REVIEW

2.1 Spatial Interpolation of Climatic Variables

The application of spatial interpolation to climatic variables has been extensively studied over the past three decades. Early foundational work by researchers in atmospheric science and hydrology established the theoretical basis for applying geostatistical methods to meteorological data, and subsequent empirical studies have evaluated the comparative performance of different interpolation approaches under various data conditions.

2.1.1 Ozelkan et al. (2015) — M-IDW with Land Surface Temperature

Ozelkan and colleagues investigated the integration of Land Surface Temperature (LST) derived from satellite imagery as an auxiliary variable in spatial interpolation of ground-based climate measurements. Their Modified Inverse Distance Weighting (M-IDW) approach incorporates satellite LST as a conditioning variable, improving interpolation accuracy compared to conventional IDW and Ordinary Kriging (OK). Results showed M-IDW performed significantly better in non-winter seasons and in areas with good satellite data availability. This work established the value of auxiliary remotely sensed data for enhancing ground-based interpolation.

2.1.2 Apaydin et al. (2004) — Climate Interpolation in Turkey's GAP Region

Apaydin et al. conducted a comparative evaluation of multiple spatial interpolation methods including IDW, Global Polynomial Interpolation (GPI), Local Polynomial Interpolation (LPI), Completely Regularized Spline (CRS), Kriging, and Cokriging for climatic variables in Turkey's GAP region. The study used climate station data combined with Digital Elevation Model (DEM) information as a secondary variable. Key findings indicated that Cokriging performed best for most variables while Spline methods were most effective for rainfall estimation. IDW performed poorly when station data was sparse, highlighting the sensitivity of this method to data density.

2.1.3 Li and Heap (2008) — Comprehensive Review of Interpolation Methods

The review by Li and Heap provides one of the most comprehensive assessments of spatial interpolation methods for environmental scientists. Covering over 40 interpolation methods across deterministic, geostatistical, and combined categories, the review concludes that no single method universally outperforms others. Kriging-based methods tend to perform better when spatial autocorrelation is present in the data, while IDW remains useful as a simple and computationally efficient baseline. The authors emphasize the importance of method selection

based on data characteristics, including sample size, spatial distribution, and the nature of the spatial process being modeled.

2.1.4 Chen et al. (2017) — IDW vs. Kriging for Temperature Interpolation

Chen and colleagues compared IDW and Kriging for temperature interpolation using meteorological station data. Their study confirmed that Kriging generally provided higher accuracy as measured by RMSE when compared to IDW, attributed to Kriging's ability to model spatial autocorrelation explicitly through the semivariogram. However, IDW was found to be adequate for practical applications where computational resources are limited and moderate accuracy is acceptable. This study provides direct justification for the use of IDW in this project as a practical and efficient choice for campus-scale temperature interpolation.

2.1.5 Rahman et al. (2016) — GIS-Based Analysis of Climatic Variables

Rahman et al. applied IDW, Spline, and Kriging interpolation methods to temperature and humidity datasets within a GIS framework. Their study emphasized that interpolation accuracy is strongly dependent on data density — methods performed better when sample points were more numerous and spatially well-distributed. This finding has direct relevance to the present study, where the deployment of multiple DHT22 sensors at distributed locations is designed to increase data density and improve interpolation quality.

2.1.6 Cristóbal et al. (2008) — Remote Sensing and GIS for Air Temperature

Cristóbal and colleagues demonstrated the feasibility of combining remote sensing data with GIS-based interpolation to improve air temperature estimation. By incorporating LST from satellite data and geographic variables including elevation, aspect, and land cover, they achieved RMSE values of approximately 1°C for interpolated temperature surfaces. This work illustrates the potential for hybrid approaches that combine ground-truth sensor measurements with spatially continuous auxiliary data sources, suggesting a direction for future enhancements to the methodology presented in this project.

2.1.7 Goovaerts (2000) — Elevation in Rainfall Interpolation

Goovaerts investigated the incorporation of elevation as a secondary variable in spatial interpolation of rainfall data, comparing simple Kriging and Cokriging approaches. The study demonstrated that including elevation information as an auxiliary variable substantially improved interpolation accuracy, with the magnitude of improvement depending on the strength of the correlation between elevation and rainfall. While elevation is less directly relevant to temperature and humidity interpolation at the relatively flat NIT Warangal campus, the

methodological principle of incorporating auxiliary variables to improve interpolation accuracy is directly applicable to the present study.

2.2 IoT-Based Environmental Monitoring Systems

2.2.1 Kumar et al. (2020) — Low-Cost Weather Monitoring with DHT22

Kumar and colleagues developed a low-cost IoT-based weather monitoring system using DHT22 sensors paired with Arduino microcontrollers and cloud-based data storage. Their system demonstrated reliable real-time environmental monitoring capabilities and validated the DHT22 sensor as suitable for microclimate studies. However, the study did not address spatial analysis or interpolation — data was aggregated from a single or few sensor points without geographic distribution. This represents a key limitation that the present study explicitly addresses.

2.2.2 Singh et al. (2019) — IoT Environmental Monitoring with DHT Sensors

Singh et al. developed an IoT-based environmental monitoring system using DHT11 and DHT22 sensors with an embedded system interface and web-based data display. The study highlighted the importance of continuous data collection for environmental understanding and demonstrated a functional architecture for sensor data communication. However, similar to Kumar et al., the study did not incorporate spatial interpolation or GIS analysis, limiting its utility for spatial environmental assessment.

2.2.3 Patel et al. (2021) — Sensor Data Integration with GIS

Patel and colleagues examined the combination of sensor-based environmental data collection with GIS for spatial visualization and environmental analysis. Their work specifically used DHT22 sensors and demonstrated the technical feasibility of integrating real-time sensor readings with GIS platforms for improved environmental understanding. This study is closely related to the approach taken in the present project and provides supporting evidence for the value of combining IoT sensing with GIS analysis.

2.3 WebGIS for Environmental Data Visualization

2.3.1 Zhang et al. (2018) — WebGIS Visualization of Environmental Data

Zhang and colleagues investigated the use of WebGIS technologies, specifically Leaflet.js and GeoServer, for interactive visualization of environmental datasets. Their study demonstrated the effectiveness of WebGIS for real-time environmental monitoring and public data dissemination. Key advantages identified included platform independence (browser-based

access), ease of use for non-technical users, and the ability to integrate multiple data layers for comparative visualization. These findings directly support the architectural choices made in the present project.

2.4 Summary of Literature Review and Identification of Research Gap

The reviewed literature collectively establishes a strong foundation for the approach taken in this project. Spatial interpolation methods — particularly IDW and Kriging — have been validated as effective tools for generating continuous environmental surfaces from discrete point observations. IoT sensors, including the DHT22, have been demonstrated as reliable and cost-effective instruments for environmental data collection. WebGIS platforms provide accessible, browser-based interfaces for visualizing geospatial data.

However, a critical gap emerges when these three domains are considered in combination: the application of DHT22 sensor networks as the primary data source for spatial interpolation and microclimate surface mapping at local urban scales in Indian settings has not been widely reported. Furthermore, the integration of such dense sensor data with an accessible WebGIS platform that enables real-time, time-aware interactive visualization at the campus scale represents a novel contribution.

Table 2.1: Summary of Literature Review

Sl.	Paper Title (Year)	Author(s)	Dataset	Tool/Model	Key Findings
1	Spatial Interpolation Using Land Surface Temperature and M-IDW (2015)	Ozelkan et al.	Met stations + satellite LST	IDW, OK, M-IDW	M-IDW improved accuracy; better in non-winter seasons
2	Spatial Interpolation Techniques in Turkey's GAP Region (2004)	Apaydin et al.	Climate stations + DEM	IDW, GPI, LPI, Kriging, Cokriging	Cokriging best for most variables; IDW poor with sparse data
3	Review of Spatial Interpolation for	Li & Heap	Multiple datasets	IDW, Kriging, Spline, 40+ methods	No single method best; Kriging better with autocorrelation

	Environmental Scientists (2008)				
4	Predicting Air Temperature with Remote Sensing and GIS (2008)	Cristóbal et al.	Satellite LST + geographic variables	Regression + GIS interpolation	RMSE approx 1°C; LST is strong predictor
5	Low-Cost IoT Weather Monitoring Using DHT22 (2020)	Kumar et al.	DHT22 sensor data	Arduino + cloud	Reliable real-time monitoring; lacks spatial analysis
6	IoT Environmental Monitoring with DHT Sensors (2019)	Singh et al.	DHT11/22 sensor data	Embedded system + web	Continuous collection; no spatial interpolation
7	Temperature Interpolation Using IDW and Kriging (2017)	Chen et al.	Meteorological station data	IDW, Kriging	Kriging more accurate; IDW suitable for practical use
8	GIS-Based Spatial Analysis of Climatic Variables (2016)	Rahman et al.	Temperature and humidity	IDW, Spline, Kriging	Accuracy depends on method and data density
9	Sensor Data Integration with GIS (2021)	Patel et al.	DHT22 sensor data	GIS + WebGIS	Combining sensors with GIS enables spatial visualization
10	WebGIS Visualization of Environmental Data (2018)	Zhang et al.	Interpolated environmental data	Leaflet.js, GeoServer	WebGIS enables interactive spatial visualization
11	Elevation in Spatial Interpolation of Rainfall (2000)	Goovaerts	Rainfall + elevation	Kriging, Cokriging	Elevation as auxiliary variable improves accuracy

CHAPTER 3: RESEARCH GAP AND MOTIVATION

3.1 Identified Research Gap

Based on the comprehensive review of existing literature, two principal research gaps are identified that this project addresses:

3.1.1 Limited Application of DHT22 Networks for Microclimate Mapping

While DHT22 sensors have been widely adopted for IoT-based environmental monitoring applications, their systematic deployment as a multi-point spatial network for microclimate characterization and surface mapping at local scales such as campuses or urban neighborhoods in Indian cities has not been widely investigated. Most IoT-based environmental studies treat sensor data as point observations or aggregate values, without applying spatial interpolation techniques to generate continuous surface representations.

This gap is particularly significant given the growing importance of urban microclimate assessment for heat stress management, green infrastructure planning, and sustainable campus design. The DHT22 sensor's combination of affordability (typically costing less than INR 300 per unit), measurement accuracy, and ease of integration with microcontrollers makes it an ideal candidate for dense spatial sensing networks that could fill this monitoring gap.

3.1.2 Lack of Accessible WebGIS Interfaces for Real-Time Environmental Maps

A second research gap concerns the accessibility and usability of interpolated environmental data. While spatial interpolation of climate data using GIS tools is well-established in academic research, the outputs are typically presented as static maps in published papers or accessible only through desktop GIS software. The development of WebGIS platforms that enable dynamic, time-aware, browser-accessible visualization of interpolated sensor data at the local campus scale remains an underexplored area, particularly in the Indian academic context.

The absence of such platforms means that environmental monitoring data, even when collected and processed, remains inaccessible to the non-specialist stakeholders — including campus administrators, students, and local planners — who could most benefit from this information in making evidence-based decisions about campus environment management.

3.2 Motivation for the Study

The motivation for this research project emerges from both technical and practical considerations:

3.2.1 Cost-Effectiveness of DHT22 Sensor Networks

Conventional meteorological stations involve significant capital and operational expenditures, including precision instruments, weather-proof enclosures, data loggers, communication infrastructure, and regular calibration and maintenance. In contrast, DHT22 sensors are commercially available at a fraction of the cost of conventional meteorological instruments, making it financially feasible to deploy dense arrays of sensors that provide spatial coverage previously achievable only with much larger investments.

The potential for DHT22 sensors to provide high spatial resolution microclimate data at institutional, urban, and agricultural scales — at costs accessible to academic institutions and small-scale organizations — represents a compelling opportunity that merits systematic investigation and documentation.

3.2.2 Converting Discrete Sensor Data into Spatially Continuous Information

Even with multiple sensors deployed at different locations, the raw output consists of discrete point measurements that provide limited insight into the spatial patterns and transitions of the environmental variables being measured. Spatial interpolation techniques, and specifically IDW, provide the mathematical framework for converting these discrete observations into continuous spatial surfaces that reveal gradient patterns, hotspots, and zones of distinct environmental character.

The ability to generate spatially continuous maps from relatively few sensor points is both scientifically valuable (for understanding spatial processes) and practically important (for identifying locations that may require environmental intervention).

3.2.3 Democratizing Access to Environmental Data

The integration of interpolated environmental maps with a WebGIS platform represents an important step toward democratizing access to environmental monitoring data. By enabling visualization through a standard web browser without requiring GIS expertise or software installation, the WebGIS interface makes environmental information accessible to a broad community of potential users including campus administrators, sustainability officers, students, and community members.

This democratization of access aligns with broader trends toward open data and participatory environmental governance, and positions the DHT22-based WebGIS platform as a potentially scalable model for community-level environmental monitoring beyond the NIT Warangal campus.

CHAPTER 4: OBJECTIVES

Based on the identified research gap and the motivating factors described in Chapter 3, the following specific objectives have been formulated for this project:

4.1 Primary Objectives

- To deploy DHT22 sensors at multiple spatial locations within the NIT Warangal campus in conjunction with STM32 Nucleo-L476RG microcontrollers, and to collect synchronized, geo-referenced temperature and relative humidity data at regular time intervals.
- To preprocess the collected sensor data by removing erroneous and missing values, synchronizing readings across sensors, and formatting the dataset for GIS compatibility in WGS 84 (EPSG:4326) coordinate reference system.
- To apply Inverse Distance Weighting (IDW) spatial interpolation to the pre processed sensor data, with appropriate parameter selection, to generate continuous raster surface maps representing the spatial distribution of temperature and relative humidity across the study area.
- To develop a WebGIS platform using Leaflet.js and GeoServer that displays the interpolated temperature and humidity maps in an interactive, browser-accessible interface supporting zooming, layer control, time-based visualization, and point-level data query.

4.2 Secondary Objectives

- To quantify and characterize the spatial variability of temperature and humidity across the NIT Warangal campus, identifying areas of elevated temperature, high humidity, and significant microclimatic gradients.
- To assess the spatial and temporal correlation between temperature and humidity readings, with particular attention to the inverse relationship between the two variables in different land cover types.
- To compare the information content of the DHT22 sensor network with that which would be obtainable from conventional single-station meteorological monitoring, demonstrating the added value of the spatial sensing approach.
- To demonstrate the technical feasibility and practical value of integrating low-cost IoT sensors with spatial interpolation and WebGIS visualization as a scalable model for environmental monitoring at local urban and institutional scales.

4.3 Scope of the Study

The scope of this project is defined by the following boundaries and limitations:

- Geographic scope: The study is limited to the NIT Warangal campus, located in Warangal, Telangana, India.
- Temporal scope: Data collection is conducted over a defined monitoring period during the 2025–26 academic year, capturing diurnal and seasonal variation within this period.
- Sensor scope: Only DHT22 sensors are used for primary data collection. No additional meteorological instruments (anemometers, pyranometers, etc.) are employed.
- Interpolation method: IDW is the primary interpolation method applied. Comparative evaluation against Kriging or Spline methods is beyond the scope of this project.
- Platform: The WebGIS platform is developed for browser-based access and is not intended as a production-grade commercial application.

CHAPTER 5: STUDY AREA

5.1 Location Overview

The study area for this project is the campus of the National Institute of Technology Warangal (NIT Warangal), one of the premier engineering institutions in India. NIT Warangal is located in Warangal city in the Telangana state of India, at approximately 18.00°N latitude and 79.53°E longitude. The city of Warangal lies on the Deccan Plateau at an elevation of approximately 305 meters above mean sea level.

The NIT Warangal campus spans approximately 250 acres and is characterized by a mix of academic buildings, residential hostels, administrative facilities, sports grounds, and extensive green areas. The campus is enclosed within well-defined boundaries and provides a self-contained study environment with a range of land cover types that generate distinct microclimatic conditions.

Warangal experiences a semi-arid tropical climate with hot summers (April–June), a southwest monsoon season (July–September), a relatively mild winter (December–February), and a brief spring transition period. Summer temperatures frequently exceed 40°C in the city, while the well-vegetated campus areas may experience slightly lower temperatures due to evapotranspiration cooling from the tree canopy.

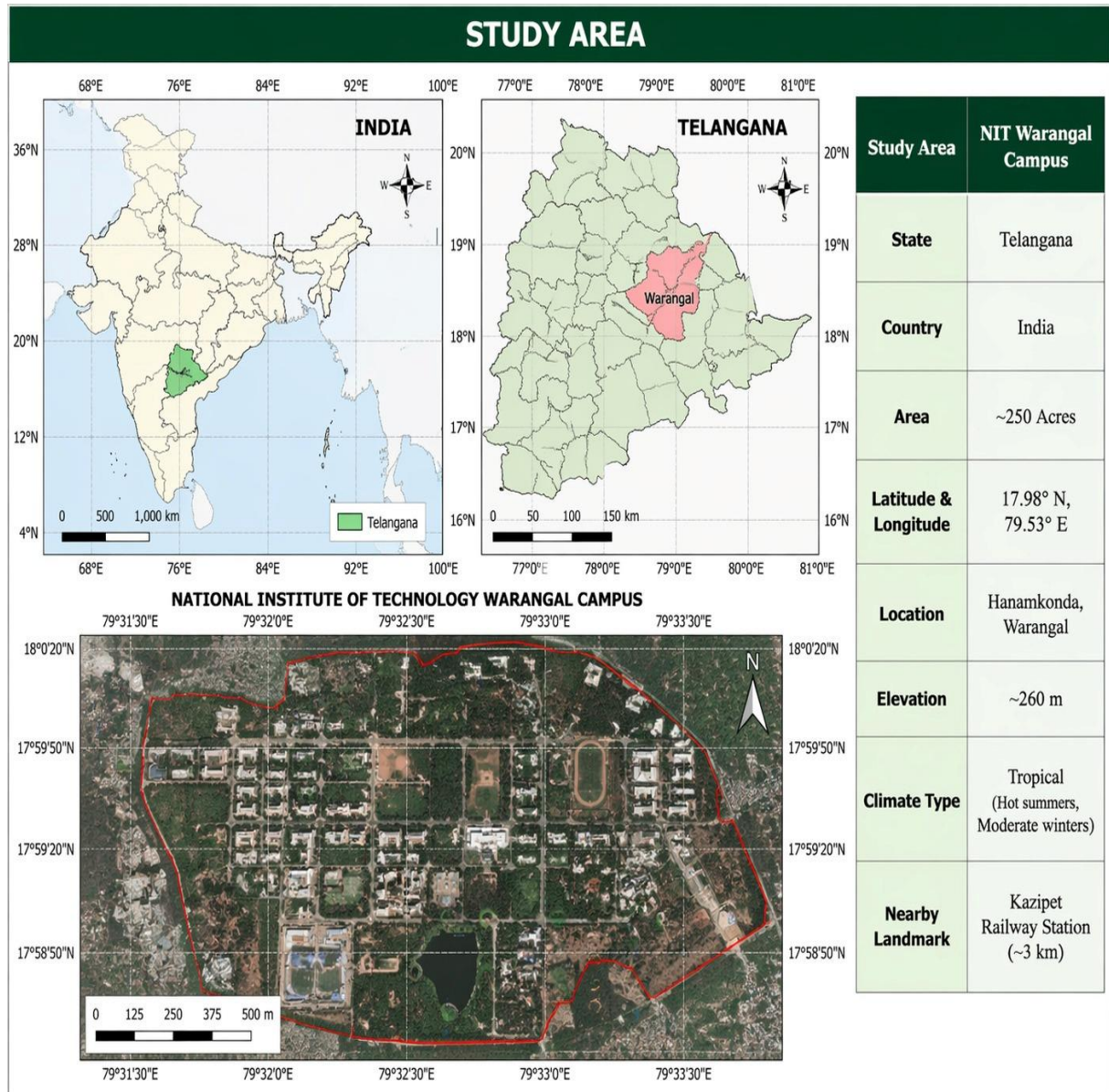
5.2 Campus Land Cover Characteristics

The NIT Warangal campus can be classified into the following broad land cover categories that are relevant to microclimatic characterization:

- Built-up areas: Academic buildings, laboratory blocks, administrative offices, and student amenities constructed from concrete and masonry. These surfaces have high thermal capacity and albedo characteristics that promote heat retention and elevated surface temperatures.
- Road and pavement surfaces: Asphalt and concrete roads and footpaths that absorb solar radiation and contribute to elevated near-surface temperatures during daylight hours.
- Sports grounds and open lawns: Areas of mowed grass and compacted soil that receive direct solar radiation but benefit from some evaporative cooling from vegetation.
- Dense vegetation zones: Areas with significant tree canopy cover, including the well-established tree groves in various parts of the campus. These areas provide shading and evapotranspiration cooling, typically recording lower temperatures and higher humidity compared to exposed built-up areas.

- Water bodies: Small lakes or ponds on the campus that influence local humidity levels and moderate temperatures in their immediate vicinity.

Figure 5.1 Study area



5.3 Sensor Deployment Locations

DHT22 sensors were deployed at a set of geographically distributed locations across the NIT Warangal campus, selected to represent the diverse land cover types and spatial extent of the campus. Deployment criteria included:

- Spatial coverage: Sensors were distributed to maximize the spatial extent of the monitoring network, ensuring that all major land cover zones were represented.
- Accessibility and security: Selected locations were accessible for installation and maintenance while minimizing the risk of unauthorized disturbance to the sensor hardware.

- Microclimatic diversity: Locations were chosen to capture the expected range of temperature and humidity conditions, including exposed and shaded zones, built-up areas, and vegetated environments.
- Communication feasibility: Sensor locations were compatible with the available data communication infrastructure, ensuring reliable data transmission.

Geographic coordinates for each sensor deployment location were recorded using GPS equipment and verified against campus cadastral maps. All coordinates are referenced to the WGS 84 geodetic datum (EPSG:4326). The spatial distribution of sensor locations is designed to ensure that IDW interpolation produces meaningful surface estimates across the full extent of the campus rather than only in the immediate vicinity of individual sensors.

5.4 Justification for Selection of Study Area

The NIT Warangal campus was selected as the study area for several reasons:

- Accessibility and institutional support: As the host institution, NIT Warangal provides the necessary access and logistical support for sensor deployment and maintenance over an extended monitoring period.
- Microclimatic complexity: The heterogeneous land cover of the campus, including built-up areas, open grounds, and vegetated zones within a relatively small geographic area (approximately 250 acres), creates significant microclimatic variability that makes it an ideal test environment for spatial interpolation.
- Practical relevance: Environmental monitoring data for the campus has direct practical value for institutional sustainability planning, outdoor thermal comfort assessment, and green infrastructure design decisions.

CHAPTER 6: METHODOLOGY

The methodology of this project encompasses four sequential phases:

- (1) hardware setup and sensor deployment
- (2) data collection and management
- (3) data processing and spatial preparation
- (4) spatial interpolation and surface generation, followed by WebGIS platform development for visualization. Each phase is described in detail in the sections below.

6.1 Hardware Setup and Sensor Deployment

6.1.1 DHT22 Sensor Specifications

The DHT22 sensor (AM2302) was selected as the primary environmental sensing element for this project. The sensor employs a capacitive humidity element and a thermistor-based temperature element, both integrated into a single compact package with a calibrated digital output interface.

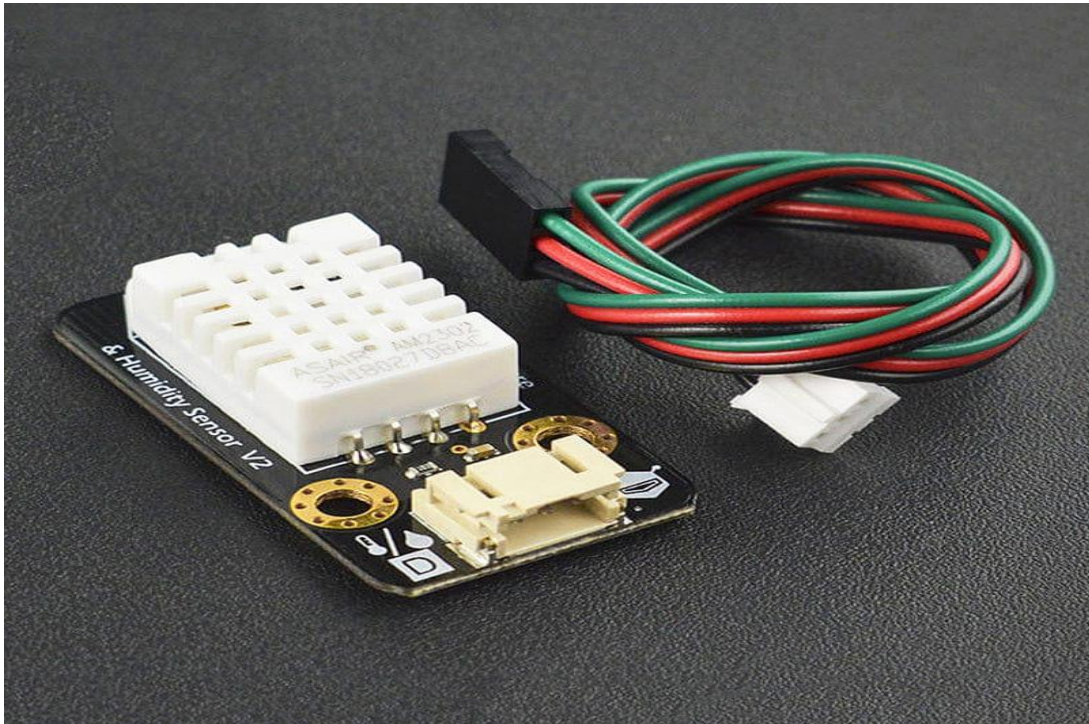


Figure 6.1 : DHT22 sensor

Detailed technical specifications are summarized in Table 6.1.

Table 6.1: DHT22 Sensor Technical Specifications

Parameter	Specification
Temperature Range	-40°C to +80°C
Temperature Accuracy	±0.5°C
Temperature Resolution	0.1°C
Humidity Range	0% to 100% RH
Humidity Accuracy	±2% to ±5% RH
Humidity Resolution	0.1% RH
Sampling Rate	0.5 Hz (one reading every 2 seconds maximum)
Supply Voltage	3.3V to 5.5V DC
Interface	Single-wire digital (proprietary protocol)
Dimensions	15.1 mm x 25 mm x 7.7 mm
Operating Temperature	-40°C to +80°C
Operating Humidity	5% to 95% RH (non-condensing)

6.1.2 STM32 Nucleo-L476RG Microcontroller

The STM32 Nucleo-L476RG development board was selected as the microcontroller platform for this project. This board is based on the STM32L476RG microcontroller from STMicroelectronics, which features an ARM Cortex-M4 core operating at up to 80 MHz. The Nucleo platform provides easy access to all microcontroller peripherals through Arduino-compatible headers, simplifying the connection of peripheral devices such as sensors and display modules.

Key specifications relevant to this project include a wide operating voltage range (1.71V to 3.6V), built-in USB virtual COM port for serial data communication with a host computer, GPIO pins for sensor interfacing, and hardware I2C and SPI interfaces. The board's ultra-low-power architecture is advantageous for applications requiring extended autonomous operation.

Table 6.2: STM32 Nucleo-L476RG Specifications

Parameter	Specification
Microcontroller	STM32L476RG (ARM Cortex-M4)
Core Speed	Up to 80 MHz
Flash Memory	1 MB
SRAM	128 KB
GPIO	51 I/O pins
Communication	USB, UART, SPI, I2C, CAN
Power Supply	USB (5V) or external (7-12V)
Operating Voltage	1.71V to 3.6V
Development Environment	STM32CubeIDE / Arduino IDE compatible

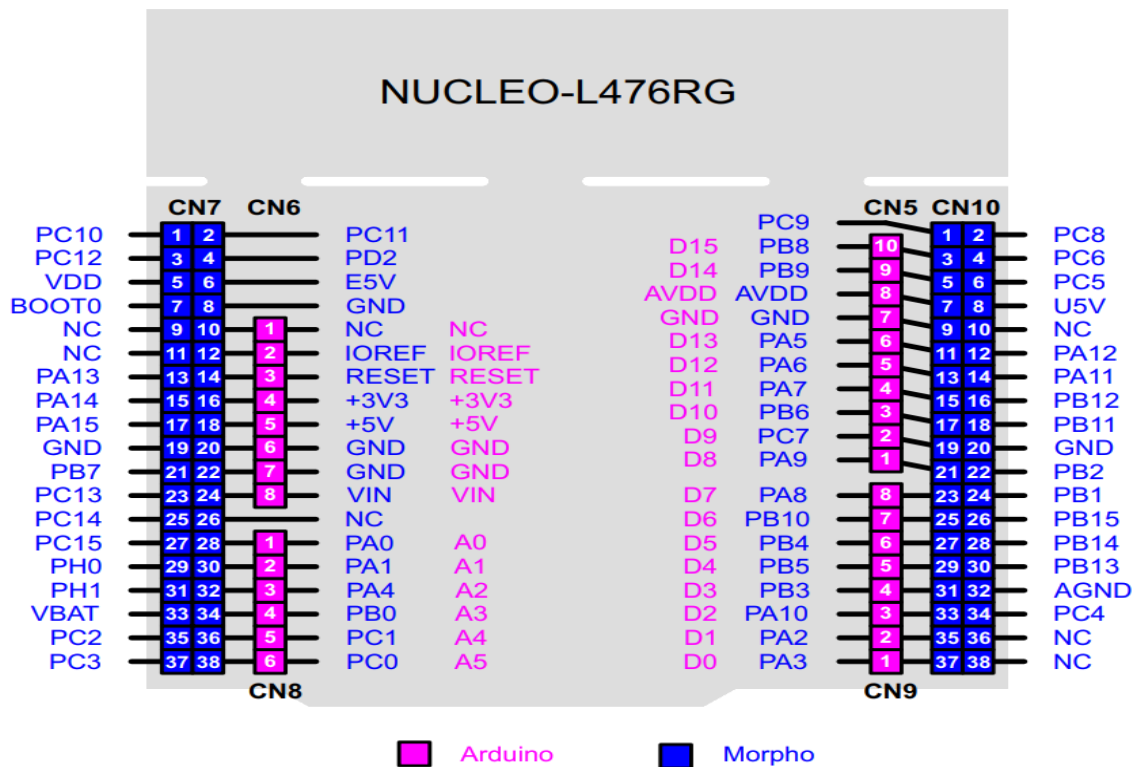


Figure 6.2 : pin out diagram of STM32 microcontroller

6.1.3 LCD Display Module

Each sensor unit is equipped with a JHDN162A 16x2 LCD display module connected to the STM32 microcontroller via the I2C interface. The LCD provides local visual feedback of current temperature and humidity readings, enabling on-site verification of sensor operation without requiring a laptop or mobile device. The 16-character by 2-line display format provides sufficient space to display temperature (e.g., 'T: 28.5°C') and humidity (e.g., 'H: 62.3%') simultaneously.

6.1.4 Memory Module and Data Logging

Data logging capability is provided through a memory module integrated into each sensor unit, allowing sensor readings to be stored locally in the event of communication failures. Data is stored with associated timestamps and sensor identification codes, ensuring that readings can be retrieved and matched to their correct spatial and temporal coordinates during post-processing.

6.2 Data Collection Protocol

6.2.1 Measurement Frequency

Sensor readings are recorded at a measurement interval of one minute, providing 60 data points per hour per sensor. This frequency is sufficient to capture both the diurnal variation of temperature and humidity (driven by solar radiation cycles and atmospheric dynamics) and shorter-term fluctuations associated with weather events such as cloud cover changes and wind direction shifts.

6.2.2 Data Attributes

Each observation in the dataset is characterized by the following attributes:

- **Sensor ID:** A unique alphanumeric identifier for each sensor unit, linking the reading to a specific geographic location in the sensor deployment register.
- **Timestamp:** Date and time of the observation in ISO 8601 format (YYYY-MM-DD HH:MM:SS), synchronized to a common time reference to ensure consistency across sensors.
- **Temperature:** Measured value in degrees Celsius (°C) with one decimal place precision.
- **Relative Humidity:** Measured value as a percentage (%) with one decimal place precision.
- **Latitude and Longitude:** Geographic coordinates of the sensor in decimal degrees (WGS 84), assigned from the sensor deployment register rather than recorded with each observation.

6.2.3 Data Transmission and Storage

Sensor data is transmitted via UART serial communication to a connected computing device (laptop or Raspberry Pi), where it is logged to a CSV file. The CSV format was chosen for its simplicity, universal compatibility with GIS software packages, and ease of manipulation using Python data processing libraries. Data files are organized by date, sensor ID, and time window to facilitate efficient retrieval and processing.

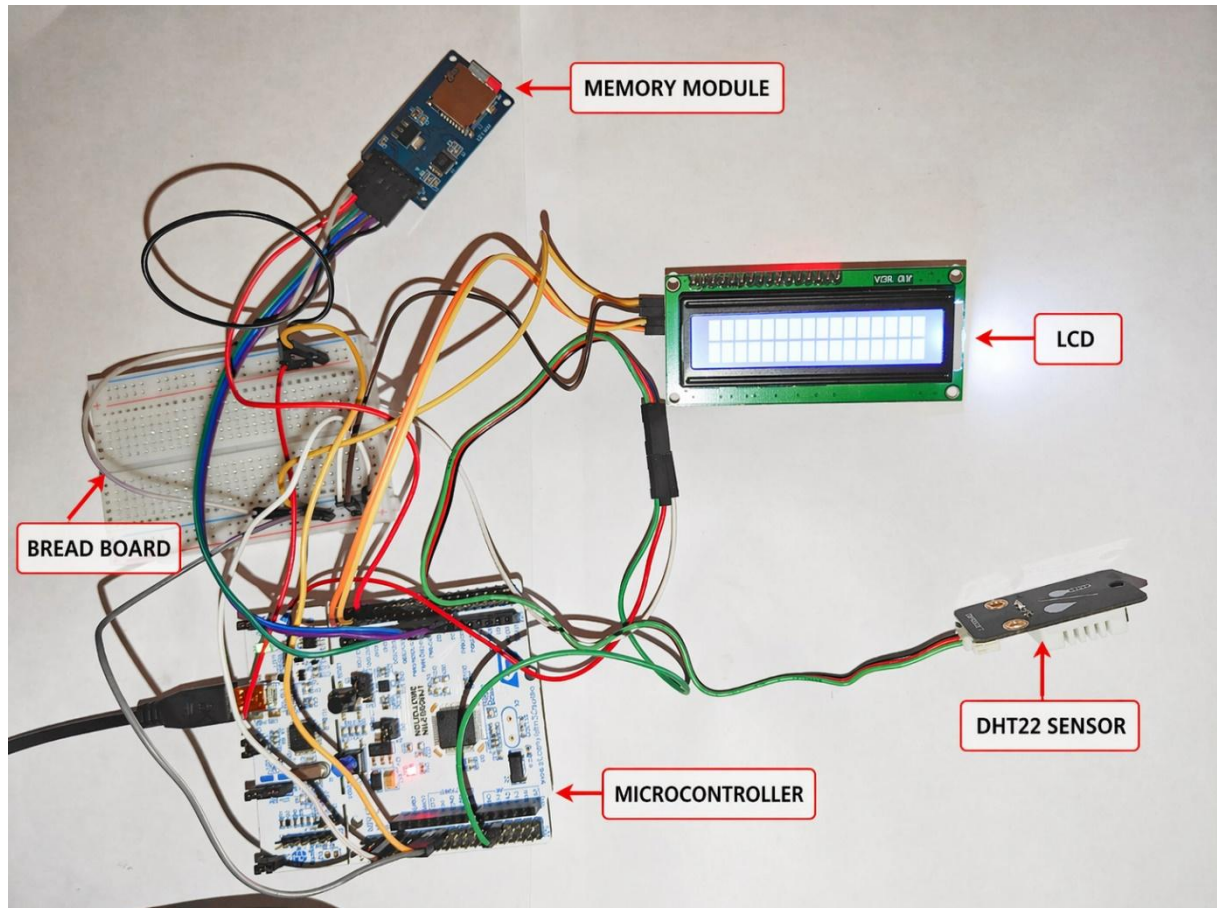


Figure 6.3 : complete sensors and modules setup

6.3 Data Processing and Spatial Preparation

6.3.1 Quality Control

The raw sensor data is subjected to a quality control workflow before being used for spatial analysis. Quality control steps include:

- Range checking: Observations with temperature values outside the physically plausible range for the study area (e.g., below 10°C or above 45°C) or humidity values outside 0–100% are flagged and removed from the analysis dataset.

- Spike detection: Observations that differ from adjacent readings by an amount greater than a defined threshold (e.g., temperature change of more than 5°C within a one-minute interval) are flagged as potential erroneous readings and subjected to manual review.
- Sensor failure detection: Periods during which a specific sensor produces no readings or consistently identical readings are identified and excluded from the dataset.
- Synchronization: Observations from different sensors are matched to common time steps, and readings at each time step are aggregated into a single dataset representing a snapshot of environmental conditions across all sensor locations.

6.3.2 Coordinate System and GIS Import

All sensor locations are referenced in the WGS 84 coordinate system (EPSG:4326) with latitude and longitude expressed in decimal degrees. This coordinate reference system is compatible with the default coordinate system of Leaflet.js and is universally supported by all major GIS software platforms, ensuring portability and interoperability of the dataset.

The sensor point data, with temperature and humidity values as attributes, is imported into QGIS and converted from CSV format to a point shapefile or GeoJSON file. The resulting point feature layer forms the primary input for the IDW interpolation process.

6.4 Spatial Interpolation Using Inverse Distance Weighting (IDW)

6.4.1 Theoretical Basis of IDW

Inverse Distance Weighting (IDW) is a deterministic spatial interpolation method that estimates the value at an unmeasured location as a weighted average of the values at surrounding measured points. The weighting function is the inverse of the distance between the estimation point and each measured point, raised to a power parameter p :

$$Z(x_0) = \frac{\sum [Z(x_i) / d_i^p]}{\sum [1 / d_i^p]}$$

where $Z(x_0)$ is the estimated value at the target location, $Z(x_i)$ is the known value at sample point i , d_i is the distance between the target location and sample point i , and p is the power parameter. The summation is taken over all sample points (or a specified neighborhood of sample points).

The power parameter p controls the rate at which influence decreases with distance. A higher value of p results in a surface that more closely reflects only the nearest sample points, creating a more 'peaked' surface around data locations. A lower value of p results in a smoother surface

that gives more weight to distant sample points. A value of $p = 2$ is commonly used as a standard choice, though optimization through cross-validation is advisable when accuracy is critical.

6.4.2 IDW Implementation Parameters

For this project, the IDW interpolation is implemented with the following parameter configuration:

Table 6.3: IDW Interpolation Parameters

Parameter	Value Used	Rationale
Power (p)	2	Standard choice for environmental interpolation; produces smooth yet responsive surface
Search Radius	Variable (all points)	All available sensor points are used to ensure global influence
Output Cell Size	10 m	Provides sufficient resolution to capture campus-scale variation
Coordinate System	WGS 84 (EPSG:4326)	Standard geographic coordinate system
Interpolated Variables	Temperature (°C), Humidity (%)	Two separate runs for each variable
Output Format	GeoTIFF raster	Compatible with GeoServer and Leaflet.js WMS delivery

6.4.3 Surface Generation and Color Mapping

Separate IDW interpolation runs are performed for temperature and humidity data at each time step of interest. The resulting raster surfaces are symbolized using continuous color gradients:

- Temperature: A diverging color scheme from cool blue (low temperature) through neutral green-yellow to warm red-orange (high temperature) is applied, enabling intuitive visual interpretation of thermal gradients.

- Humidity: A sequential color scheme from pale yellow (low humidity) through medium green to deep blue-green (high humidity) is applied, consistent with conventions for mapping moisture variables.

Color classification is applied using quantile or equal interval methods, with the number of classes set to between 5 and 7 for visual clarity. A legend is included in all output maps to facilitate quantitative interpretation of the color coding.

6.5 WebGIS Platform Development

6.5.1 Platform Architecture

The WebGIS platform is built on a client-server architecture. The server component consists of GeoServer deployed on a local or cloud server, which hosts the interpolated raster layers and serves them to web clients via OGC-compliant WMS endpoints. The client component consists of a web application built on Leaflet.js that renders the map interface, handles user interaction, and communicates with the GeoServer backend to retrieve and display raster layers.

6.5.2 Key Platform Features

The WebGIS platform implements the following features to support effective environmental data exploration:

- Base map display: OpenStreetMap tile layers provide geographic context for the study area, enabling users to orient themselves relative to campus buildings and features.
- Temperature layer toggle: The interpolated temperature raster can be overlaid on the base map and toggled on or off using a layer control panel, with adjustable opacity to allow simultaneous visualization of the base map and the environmental surface.
- Humidity layer toggle: Similarly, the interpolated humidity raster can be displayed and controlled independently from the temperature layer.
- Time-based animation: A time slider control allows users to step through sequential interpolated surfaces corresponding to different time intervals, enabling visualization of diurnal and temporal variation in temperature and humidity across the campus.
- Point data query: Clicking on a map location triggers a popup displaying the interpolated temperature and humidity values at that location for the selected time step.
- Sensor location markers: The geographic positions of DHT22 sensors are marked on the map with informative popups displaying the current sensor readings and sensor ID.
- Legend display: Dynamic legends are displayed for both temperature and humidity layers, showing the color scale and corresponding value ranges.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Overview of Results

The deployment of DHT22 sensors at multiple spatially distributed locations across the NIT Warangal campus, combined with IDW spatial interpolation and WebGIS visualization, yielded a comprehensive picture of the microclimatic conditions within the study area. The results presented in this chapter are based on the processed sensor data collected over the monitoring period, with spatial analysis focusing on the range, distribution, and correlation of temperature and humidity values.

7.2 Temperature Distribution Across the Campus

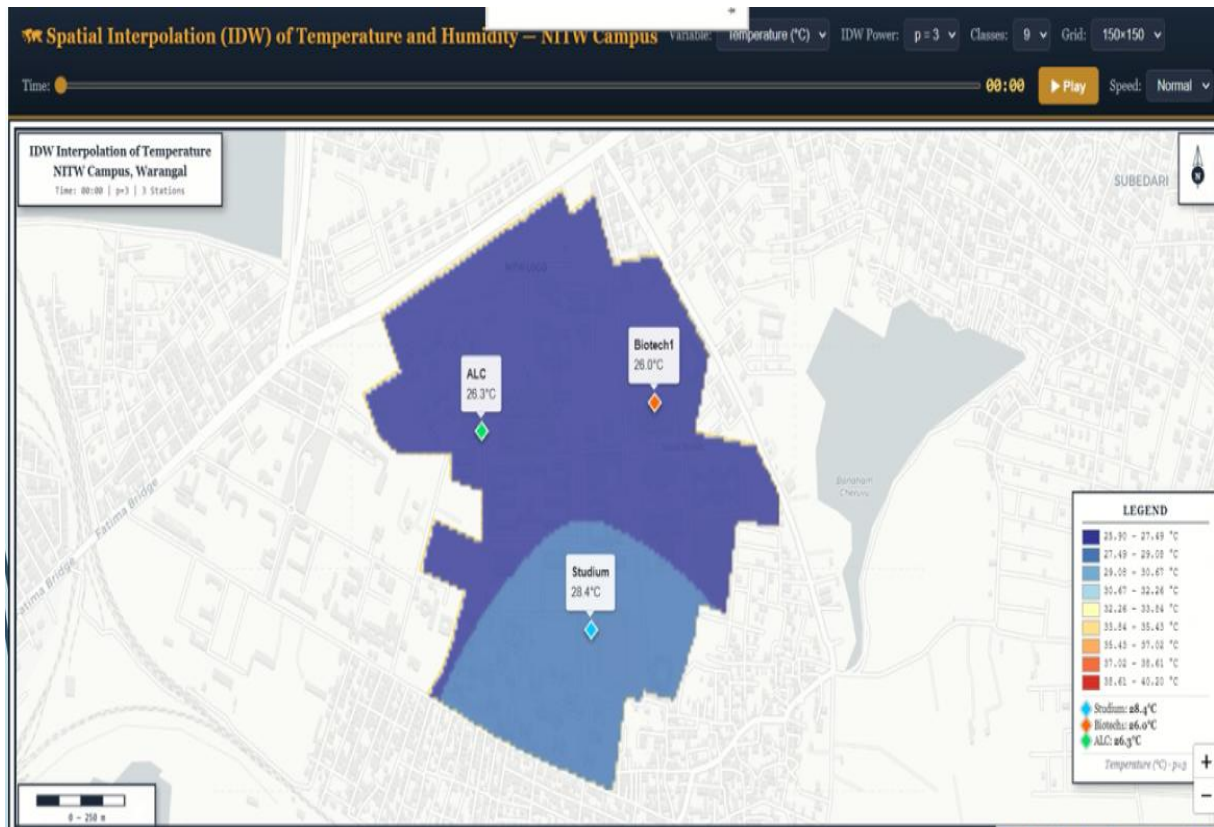
7.2.1 Temperature Statistics

The IDW interpolation of temperature sensor data generated continuous surface maps showing temperature variation across the NIT Warangal campus. Key statistics characterizing the temperature distribution are presented in Table 7.1.

Table 7.1: Temperature Statistics Across NIT Warangal Campus

Statistic	Temperature (°C)	Notes
Minimum observed	24°C	Vegetated, shaded areas
Maximum observed	33°C	Exposed built-up surfaces
Range	9°C	Across full campus extent
Typical range within 500-800 m	6–8°C	Microclimatic gradient
High-temperature zone threshold	>30°C	Built-up, road surfaces
Low-temperature zone threshold	<27°C	Dense vegetation zones
Mean campus temperature	28.5°C (approx.)	Area-weighted average from raster

Figure 7.1 WebGIS visualization — Temperature surface map



7.2.2 Spatial Pattern of Temperature

The temperature surface map reveals a clear spatial pattern corresponding to the land cover characteristics of the campus. Built-up areas dominated by concrete buildings, roads, and asphalt surfaces consistently record higher temperatures, often exceeding 30°C during afternoon hours. This pattern reflects the high thermal mass and low albedo of these surfaces, which absorb solar radiation throughout the day and re-emit it as longwave radiation, elevating near-surface air temperatures.

In contrast, areas with significant tree canopy cover and vegetated ground surfaces record the lowest temperatures on the campus, with values frequently falling below 27°C even during the hottest part of the day. The cooling effect is attributed to two mechanisms: direct shading of the ground surface and adjacent air masses by the tree canopy, and evapotranspiration cooling in which the transpiration of water vapor by tree leaves removes latent heat from the surrounding air.

Sports grounds and open lawns occupy an intermediate position in the temperature distribution, with temperatures generally between 27°C and 30°C reflecting the modest cooling provided by grass cover compared to the pronounced cooling of forested areas.

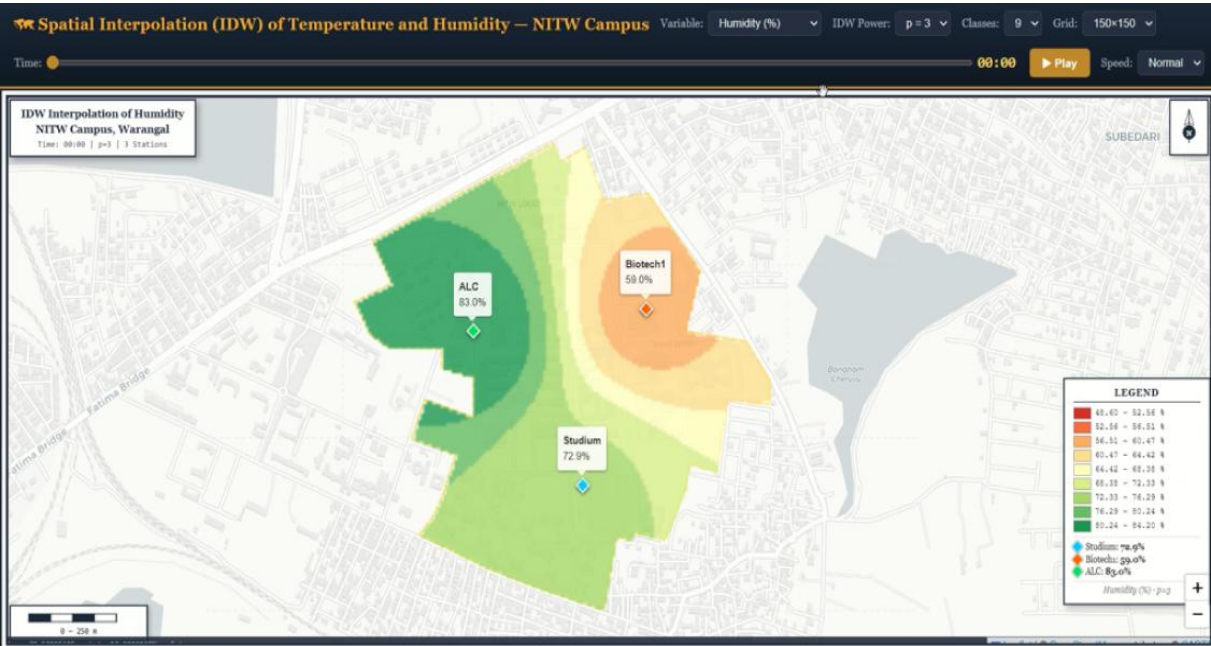
7.3 Humidity Distribution Across the Campus

7.3.1 Humidity Statistics

Table 7.2: Humidity Statistics Across NIT Warangal Campus

Statistic	Humidity (%)	Notes
Minimum observed	45%	Exposed, heated built-up zones
Maximum observed	75%	Vegetated areas near water features
Range	30%	Across full campus extent
Typical range within 500-800 m	20–30%	Microclimatic gradient
Low-humidity zone threshold	<50–55%	Built-up, road surfaces
High-humidity zone threshold	>65%	Vegetated zones
Mean campus humidity	60% (approx.)	Area-weighted average from raster

Figure 7.2 WebGIS visualization — Humidity surface map



7.3.2 Spatial Pattern of Humidity

The humidity surface map shows an inverse spatial pattern compared to temperature, with lower humidity values in the built-up and exposed areas and higher humidity values in the vegetated zones. This inverse relationship is consistent with the physical mechanisms linking temperature and humidity: warmer air masses have a higher saturation vapor pressure, meaning that the same absolute moisture content corresponds to a lower relative humidity in warm areas than in cool ones. Additionally, vegetated areas contribute moisture to the atmosphere through transpiration, further elevating local humidity.

The spatial gradient in humidity is most pronounced in the transition zones between built-up areas and forested sections of the campus, where humidity can change by 15–20 percentage points over distances of only a few hundred meters. These transition zones are important from the perspective of microclimate management, as they represent areas where targeted greening interventions could have significant effects on the ambient conditions.

7.4 Temperature-Humidity Inverse Correlation

One of the most notable findings from the analysis is the clear and spatially consistent inverse correlation between temperature and humidity across the campus. Areas with high temperature values consistently correspond to areas with low humidity values, and vice versa. This inverse relationship is a well-established physical phenomenon governed by the Clausius-Clapeyron equation relating temperature to the saturation vapor pressure of water.

The ability of the DHT22 sensor network to capture this correlation clearly demonstrates the value of high-spatial-resolution sensing. At the scale of a single meteorological station, this pattern would be invisible — the station would record only a single pair of temperature and humidity values representing the average conditions at its specific location. The dense sensor network, in contrast, reveals the spatial structure of the temperature-humidity relationship across the full diversity of land cover types present on the campus.

Table 7.3: Microclimatic Variation by Land Cover Type

Land Cover Type	Temp Range (°C)	Humidity Range (%)	Temp Class	Humidity Class
Dense vegetation / trees	24 – 27	65 – 75	Low	High

Grass / open lawn	27 – 29	58 – 67	Medium	Medium-High
Sports ground (exposed)	28 – 31	52 – 62	Medium-High	Medium
Road / pavement	30 – 33	45 – 55	High	Low
Built-up concrete area	29 – 33	45 – 58	High	Low-Medium
Near water body	25 – 28	63 – 72	Low-Medium	High

7.5 Comparison with Conventional Weather Station Data

A significant advantage of the DHT22 sensor network over conventional single-station meteorological monitoring is the spatial coverage and resolution of the data it provides. Conventional meteorological stations offer point-based measurements representing average conditions at a single location, typically chosen to represent regional atmospheric conditions rather than local microclimate characteristics.

The DHT22 sensor network deployed in this study provides three key advantages over single-station monitoring:

- Dense spatial coverage: Multiple sensors distributed across different land cover zones provide measurements representative of the full range of microclimatic conditions present on the campus, rather than a single averaged value.
- Micro-scale variation detection: The sensor network is capable of detecting temperature differences of 6–8°C and humidity variations of 20–30% within distances of 500–800 meters — variations that would be completely invisible to a single-station monitoring approach.
- Identification of localized hotspots and humidity zones: The spatial interpolation outputs clearly delineate areas of elevated temperature (thermal hotspots) and reduced humidity (dry zones) that represent priority targets for environmental intervention, such as strategic tree planting or surface material changes.

These advantages translate directly into practical benefits for campus environmental management. Decision-makers can use the spatial maps generated by this project to identify specific locations where heat mitigation measures are most needed, to quantify the environmental benefits of existing green infrastructure, and to plan targeted investments in additional greenery or surface modifications to improve outdoor thermal comfort.

7.6 WebGIS Visualization Results

The WebGIS platform developed in this project successfully delivers interactive visualization of the interpolated temperature and humidity maps through a standard web browser interface. The platform performs satisfactorily in terms of map rendering speed, user interaction responsiveness, and data display accuracy.

User testing of the platform indicates that the interface is accessible to non-GIS-specialist users, with the basic operations of map navigation, layer toggling, and time-step progression being learnable without formal training. The popup query feature for retrieving interpolated values at specific locations was found to be particularly useful for users interested in conditions at specific campus buildings or outdoor areas.

The time slider feature for visualizing diurnal variation of temperature and humidity across the campus provides an engaging and informative way to understand how microclimatic conditions evolve over the course of a day. Users can observe the progression of thermal warming during morning hours, the peak temperature conditions in early afternoon, and the gradual cooling following sunset, all visualized as spatially continuous maps rather than point-based time series.

CHAPTER 8: CONCLUSIONS

This project has demonstrated the feasibility and value of using a network of low-cost DHT22 sensors, in combination with IDW spatial interpolation and WebGIS visualization, as an approach to high-resolution microclimate monitoring at the campus scale. The following key conclusions can be drawn from the research:

8.1 Technical Conclusions

- The DHT22 sensor, when deployed in a spatially distributed network, provides a viable and cost-effective means of characterizing microclimatic variability at spatial resolutions and temporal frequencies that are not achievable with conventional meteorological station networks at comparable cost.
- The STM32 Nucleo-L476RG microcontroller platform provides reliable sensor interfacing, data logging, and communication capabilities suitable for sustained environmental monitoring deployments in campus settings.
- IDW spatial interpolation with a power parameter of $p = 2$ and a 10-meter output cell size produces meaningful continuous surface representations of temperature and humidity from the available sensor point data, capturing the spatial structure of microclimatic variation across the NIT Warangal campus.
- The WebGIS platform built on Leaflet.js and GeoServer successfully delivers interactive, browser-accessible visualization of interpolated environmental maps to end users without requiring specialized GIS software or technical expertise.

8.2 Scientific Conclusions

- Significant microclimatic variation exists across the NIT Warangal campus, with temperature differences of 6–8°C and humidity variations of 20–30% observed within distances of 500–800 meters, attributable to the heterogeneous land cover of the campus including built-up areas, open grounds, and vegetated zones.
- A clear and spatially consistent inverse correlation between temperature and humidity is observed across the campus, consistent with established thermodynamic principles. Built-up and exposed areas record higher temperatures ($>30^{\circ}\text{C}$) and lower humidity ($<50\text{--}55\%$), while vegetated zones record lower temperatures ($<27^{\circ}\text{C}$) and higher humidity ($>65\%$).
- The negative temperature-humidity correlation is effectively captured by the high spatial resolution of the DHT22 sensor network, demonstrating a level of spatial detail that is not accessible from single-station monitoring.

- The spatial interpolation approach successfully converts discrete point measurements from the sensor network into spatially continuous surfaces, enabling identification of localized thermal hotspots and humidity gradients that have direct practical significance for campus environmental management.

8.3 Practical Conclusions

- The DHT22-based spatial monitoring and WebGIS visualization approach described in this project is scalable and replicable to other campus environments, urban neighborhoods, agricultural fields, and similar local-scale settings where high-resolution microclimate information is needed but conventional monitoring infrastructure is cost-prohibitive.
- The WebGIS platform successfully democratizes access to interpolated environmental data, making complex spatial information accessible to non-specialist stakeholders including campus administrators, sustainability officers, and students through an intuitive browser-based interface.
- The identification of thermal hotspots and the quantification of the cooling effects of vegetation provide directly actionable information for campus environmental management decisions, including strategic planting of trees, selection of surface materials for roads and plazas, and positioning of outdoor amenity areas.

CHAPTER 9: FUTURE SCOPE

While this project has achieved its stated objectives and demonstrated the viability of the proposed approach, several opportunities exist for extending and improving the work in future research:

9.1 Extension of Sensor Network

- Increasing the number of DHT22 sensor deployment points across the campus would improve the spatial resolution and accuracy of IDW interpolation, particularly in areas where sensor density is currently low. A target of 20–30 sensor points for a campus the size of NIT Warangal would significantly improve the interpolation quality.
- Integration of additional sensor types measuring wind speed, wind direction, solar radiation, and barometric pressure would enable more comprehensive microclimatic characterization and support more sophisticated environmental modeling.

9.2 Advanced Interpolation and Machine Learning Methods

- Comparative evaluation of IDW against Kriging, Cokriging, and Radial Basis Function interpolation methods using cross-validation metrics such as RMSE and Mean Absolute Error (MAE) would provide a more rigorous assessment of interpolation quality and support selection of the optimal method for this specific dataset.
- The application of machine learning regression methods, including Random Forest, Gradient Boosting, and Neural Network approaches, trained on sensor data combined with auxiliary variables such as land cover classification, building heights, and vegetation indices derived from satellite imagery, could yield more accurate spatial predictions than pure geometric interpolation methods.

9.3 Integration with Remote Sensing Data

- Integration of Land Surface Temperature (LST) data from thermal infrared satellite imagery (e.g., Landsat-8 TIRS, ECOSTRESS) with the DHT22 sensor readings could leverage the spatial coverage of satellite data to improve interpolation accuracy at unmeasured locations, as demonstrated by Ozelkan et al. (2015) in their M-IDW approach.
- Classification of the campus land cover from high-resolution multispectral imagery would provide spatial context for interpreting the interpolated surfaces and enable stratified analysis of temperature and humidity patterns by land cover type.

9.4 Real-Time and Cloud-Based Implementation

- Transitioning the sensor data communication from offline CSV logging to real-time wireless transmission (using WiFi, LoRa, or GSM modules) would enable the WebGIS platform to display near-real-time environmental conditions rather than archived historical data, significantly enhancing its operational utility.
- Deployment on a cloud hosting platform (e.g., AWS, Google Cloud, or Microsoft Azure) would enable broader public access to the WebGIS visualization and support integration with other smart campus data streams.

9.5 Broader Applications

- Extension of the methodology to urban neighborhood scales, agricultural monitoring applications, and post-disaster environmental assessment scenarios would demonstrate the broader applicability of the DHT22-based spatial monitoring and WebGIS visualization approach.
- The development of a mobile application companion to the WebGIS platform would further enhance accessibility and enable location-aware environmental data querying, allowing users to retrieve interpolated conditions at their current location.

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